

# Suga' — A Live Virtual Dance Performance

VALENCIA JAMES, Independent, USA

THOMAS WESTER, Independent, USA

SIMON BOAS, Midgray, USA

HOLLY NEWLANDS, Independent, USA

MARIN VESELY, Independent, USA

SANDRINE MALARY, Independent, USA

TERRI AYANNA WRIGHT, Independent, USA



## ACM Reference Format:

Valencia James, Thomas Wester, Simon Boas, Holly Newlands, Marin Vesely, Sandrine Malary, and Terri Ayanna Wright. 2021. Suga' — A Live Virtual Dance Performance. In *Proceedings of the ACM SIGGRAPH'21 Art Gallery, August 09-13, 2021, Virtual Event, USA*. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3450507.3457436>

## PROJECT URL

<https://volumetricperformance.com>

---

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

*SIGGRAPH'21 Art Gallery, August 09-13, 2021, Virtual Event, USA*

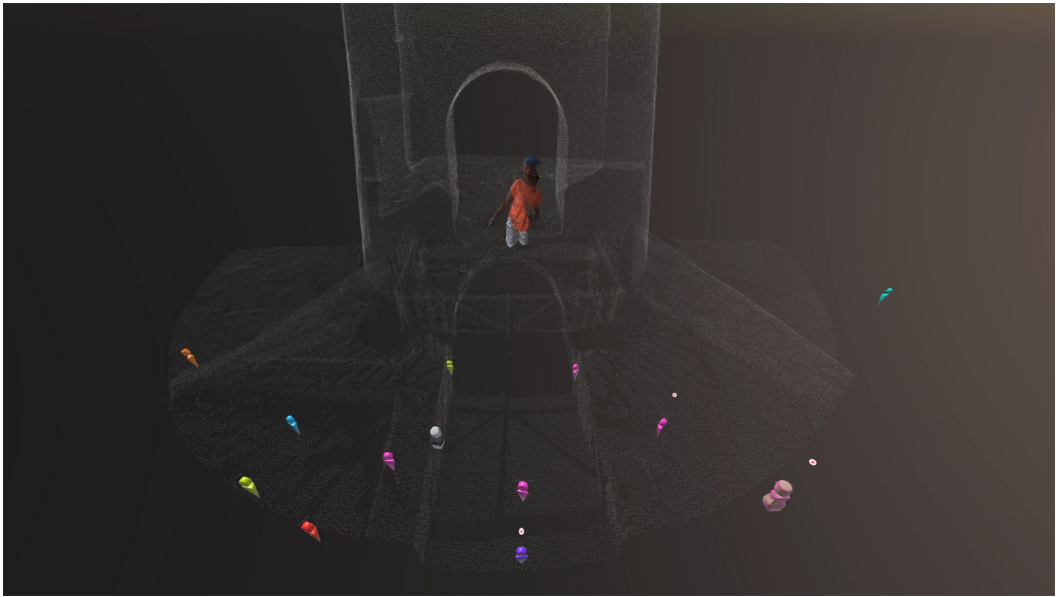
© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8360-8/21/08.

<https://doi.org/10.1145/3450507.3457436>

## DESCRIPTION OF ARTWORK

*Suga'* is a live volumetric dance performance experience presented in Mozilla Hubs' social virtual reality space. The performer appears as a live-streamed 3D video within a photogrammetry scan of the Annaberg Sugar Mill Ruins in the U.S. Virgin Islands, when in fact she is performing from her living room. She traces the point cloud rendering of the mill with her movements as if preparing the space for ritual, recalling events that happened there. Through movement and soundscape, the performer tells the story of her own deep connection and mixed relationship to similar sites as a Black woman growing up in the Caribbean.



The performance weaves together dance, family history, and 3D cultural heritage to explore virtual environments as a possible site for healing and to reclaim spaces that were historically filled with pain and injustice. It raises questions around Black femme representation in theatre and digital spaces. The work seeks to reimagine virtual environments as a possible site for what the scholar and Vodun priestess Dowoti Desir describes as spatial justice and African descendants' right to collective memory of their struggle and resistance.

The work is a bold reimagining of immersive webspace as a venue for live dance performance and communal art experience. The audience experience starts by choosing from a selection of avatars to embody while inside the performance space. A collective movement orientation exercise then follows in order to help familiarize the audience with movement in 3D space. The live, dynamic connection between audience and performer found in post-modern in-person performances is recreated in virtual space by allowing audience members to move around, and change their points of view in relation to the performer throughout the performance. A sense of intimacy and liveness is created by the consequent interplay of audience movement and direct interaction with the performer via webcam during the culminating question-and-answer session.

*Suga'* pioneers innovative advancements in live volumetric capture of dance performance for online immersive webspace. The work was developed as a research pilot performance of the *Volumetric Performance Toolbox*. Volumetric Performance Toolbox is the collaborative project of Valencia James, Glowbox, and Sorob Louie. It responds to the dire need created by theatre closures



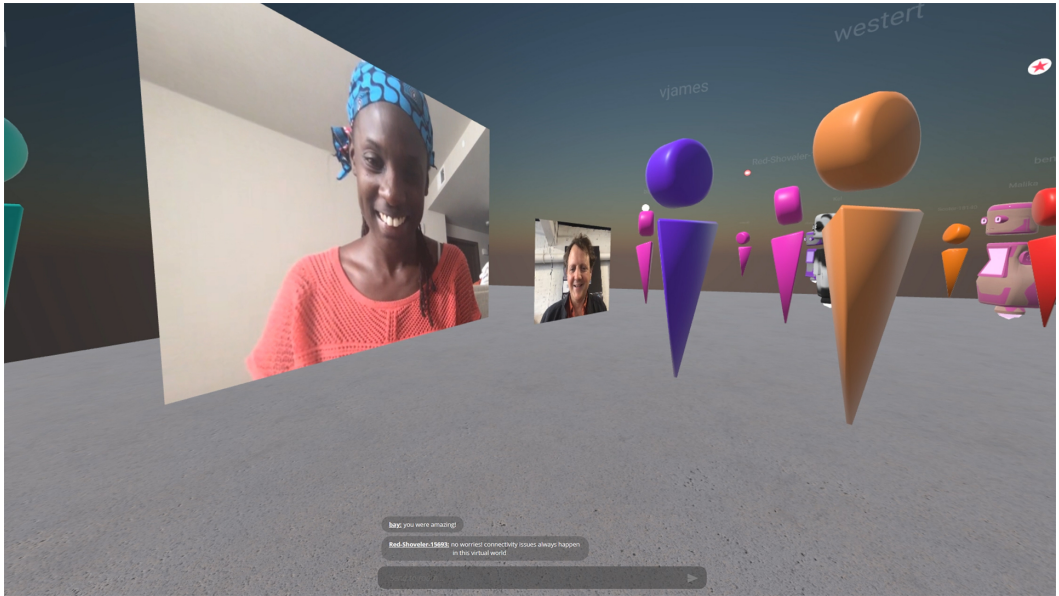
due to the global pandemic by envisioning live online 3D dance performance as a new way for artists to create and perform from their own living spaces and audiences to communally experience art, using minimal equipment and open source software.

The toolbox is an open-source prototype for volumetric dance performance in Mozilla Hubs. It consists of a low-cost hardware kit and custom software that allows movement artists to capture and stream their performances in real-time from their own living spaces, using minimal equipment. The performance can be accessed via web browser across most devices and is not limited to virtual reality headsets.

Our starting question was: How can there be new dance creation and performance in the time of COVID? In the crisis faced by thousands of theaters and dance companies, performances have been cancelled and the creation of new work halted. *Suga'* is the first performance in an ongoing effort to give artists access to tools that allow them to express themselves in live, virtual space. With the future of large social gatherings uncertain, this project helps performers create their own virtual spaces that audiences can access with no special equipment beyond a web browser.

We are particularly interested in the critical questions raised by live performance in immersive webspace. Who builds the tools of immersive technologies and who is it created for? How might virtual spaces be reclaimed by those communities that are often left out of these spaces or subjected to harmful representation or exploitative cultural appropriation when included? How might live performance in immersive webspace be used for decolonial strategies that affect real-world change and justice?

Our mission is to help movement artists create their own live volumetric performances by providing low cost, lightweight equipment, tutorials and building a community of creators. We are beginning this process by running a pilot artist residency for making volumetric performance with a cohort of seven multidisciplinary artists from BIPOC, LGBTQ, and disabled communities coming together to envision a decolonized, decentralized future for online performance.



## ARTIST INFORMATION

**Valencia James** is a Barbadian freelance performer, maker, and researcher interested in the intersection between dance, theatre, technology, and activism. She believes in the power of the arts to inspire change. Valencia's work explores how emerging technologies like machine learning and computer vision might enhance creativity in her contemporary dance practice and vice-versa. She has presented her work at several international forums such as the 2015 International Joint Conference on Artificial Intelligence in Buenos Aires and TEDxGoteborg. She has been a 2020 Rapid Response Fellow at Eyebeam, where she co-founded the Volumetric Performance Toolbox, a project that responds to the global pandemic by envisioning live online 3D dance performance as a new way for artists to connect to audiences from their living spaces using minimal equipment. Currently she is a Spring 2021 Remote Resident for Open-Source Software Art Tools at Carnegie Mellon University. <http://valenciajames.com>

**Thomas Wester** co-founded Glowbox, a spatial interaction lab based in Portland, Oregon. Through Glowbox, he builds shared reality experiences that blend physical and digital to connect people to ideas, nature, and each other. He has worked with The Royal Shakespeare Company, Library of Congress, Hermès, Microsoft, Nike, MoMA, MFA Boston, and exhibited at Tribeca and Sundance. <http://thomaswester.com>

**Simon Boas** investigates technology, culture, and aesthetics through socially engaged art and design, often working under the collective moniker Midgray. His work has been exhibited internationally through SIGGRAPH, Eyebeam, the Institute of Network Cultures, Neural, and Digicult. His art and research have received awards from the Oregon Arts Commission and the University of California. He holds a M.F.A. degree from the University of California, Santa Cruz, and is currently based in Portland, Oregon, where he works as a creative technologist and educator. <http://www.midgray.com>

**Holly Newlands** makes games and other playful experiences. Their work has been exhibited at festivals including Currents Virtual, A MAZE. / BERLIN, XOXO, and IndieCade. Holly recently graduated from the University of Oregon's Art & Technology program with a B.F.A. They currently reside in Portland, Oregon, where they work at Glowbox as a technical artist and developer. <http://thnewlands.com>

**Marin Vesely** is a Los Angeles-based artist and designer with a passion for technology, activism, and collaboration. Marin's work has been exhibited internationally in Brazil, Norway, and the United States. <https://www.marinvesely.com>

**Sandrine Malary** was born and raised in Port-au-Prince, and is part of a new generation of Haitian artists who maintain close ties to their motherland and its culture. She has a degree in Psychology from the George Washington University, which she uses to relate to her community and to parent. She is a mother of 4 children. Two home-birthed in Ayiti and two in Oakland, CA. With self-sustenance as a life approach, Sandrine has homeschooled her kids, built a home garden and created home-based on-line businesses as an artist and educator. Keeping her roots strong, she dances Haitian folklore, teaches Kreyol and her art is very inspired by Vodou. Much of her focus is to uplift, not just Ayiti, but women, black people, and our planet. Sandrine advocates for green, sustainable living with a focus on health and wellness. You can often find her walking local trails with her group "Sistas on the Trail," creating therapeutic copper jewelry for [www.houseofalouba.com](http://www.houseofalouba.com), or working on her upcoming vegan cookbook.

**Terri Ayanna Wright** hails from Baton Rouge, Louisiana by way of Houston, Texas, and has called New York City her home since 2011. She is an honors graduate of the Ailey/Fordham BFA Program, with a major in dance and minor in computer science. Immediately after graduating, Ms. Wright toured internationally as a company member with Ailey II, where she was completely enthralled by the wide-range of live production teams, theaters, and audiences from city to city. She has since been granted opportunities to choreograph and teach all over the world, most notably as an Ailey Extension Instructor in Italy for Orsalina28's Summer Intensive. After performing with Carolyn Dorfman Dance for two seasons, Ms. Wright joined The Met Opera as an ensemble dancer in Porgy & Bess, choreographed by Camille A. Brown. Through this experience, she began exploring the many facets of live production beyond the performance act itself. She is a recent honors graduate of Parsons with an MFA in Design & Technology, which has allowed her to expand her storytelling capabilities through the marriage of dance, digital video, and visual effects.